5	(a)	The current in the copper and tungsten filament wires is the same because both wires are connected in series.		[1]
	(b)	*Resistivity*, $\rho = \dfrac{AR}{L}$ $= \dfrac{\pi(0.020\times10^{-3})^2\times300}{4(1.5)} = 6.28\times10^{-8}\ \Omega\ \text{m}$		[1] [1]
	(c)	(i)	Since current is the same in both wires, using $I = neAv$ where I: current, n:number density, A: cross-sectional area, v: drift velocity of electrons Hence $neAv$ for tungsten = $neAv$ for copper $v_{tungsten} = \dfrac{n_{Cu}A_{Cu}v_{Cu}}{n_{tungsten}A_{tungsten}} = \dfrac{(8.49\times10^{28})(1.5)^2(0.021\times10^{-3})}{(3.4\times10^{28})(0.020)^2}$ $= 0.295\ \text{m s}^{-1}$ OR $0.30\ \text{m s}^{-1}$	[1] [1] [1]
		(ii)	The electrons in the tungsten filament have a significantly higher drift velocity. This implies that they collide with the positive lattice ions at a much higher speed more frequently (more collisions per unit area). They impart more kinetic energy to the lattice ions at the same time, such that the ions vibrate more vigorously and obstruct the flow of electrons. Since the tungsten filament gains thermal energy at a significantly higher rate and subsequently heats up, the filament of the lamp gets hot but the copper leads stay relatively cold.	[1] [1]
		(iii)	If the diameter of the wire is doubled, the cross-sectional area of the wire will be quadrupled. The resistance $\left(R = \dfrac{\rho l}{A}\right)$ is thus quartered. Current $\left(I = \dfrac{V}{R}\right)$ will be quadrupled. Electron number density is the same for the same material, the drift velocity of electrons in the tungsten wire ($v = I/nqA$) is unchanged.	[1] [1]